

WEATHERGPT

Problem Statement ID – SIH26068

Problem Statement Title – Weather GPT:

Conversational AI for Weather Forecasting, Alerts,
and Climate Information

Theme – Disaster Management

PS Category – Software

Team Name – CIPHER X



The idea: turn official weather data into clear answers and useful next steps.

01 Hard-to-understand forecasts

Government forecasts often use technical terms that are difficult for everyday users to understand.

02 Unclear local warnings

Critical weather alerts may not reach people in a clear, simple and regional-language format.

03 Need for actionable advice

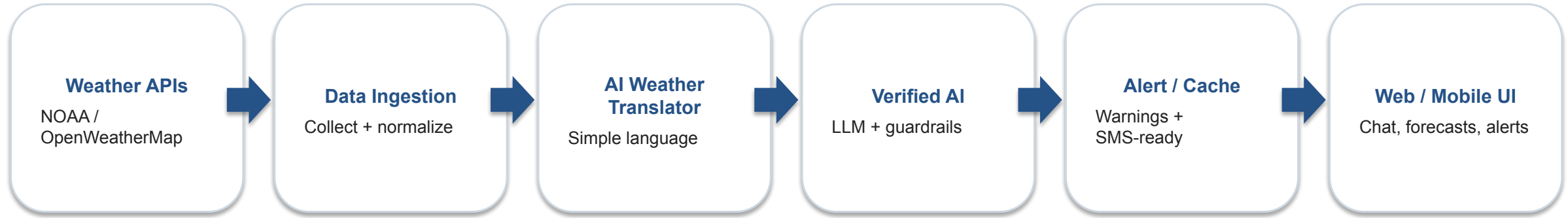
People need to know what the forecast means for them and what they should do next—not just rainfall percentages.

Our difference: explain the forecast clearly, connect it to the user's situation, and keep the information grounded in trusted weather data.

SYSTEM ARCHITECTURE



- **How the website works: ask → fetch trusted data → simplify → verify → alert → display**



APP / WEBSITE FLOW

- User enters a question or selects a location.
- System retrieves forecast, warning, radar and climate data.
- Different weather formats are normalized into one structure.
- The translator converts technical data into simple language.

RELIABILITY + DELIVERY

- Verified AI answers only from retrieved weather data and official warnings.
- Guardrails keep the response grounded in the source data.
- Critical alerts are cached for weak connectivity.
- Web/mobile UI delivers forecasts, alerts and climate information.

TECHNICAL FEASIBILITY

- Uses existing weather APIs such as NOAA and OpenWeatherMap.
- LLM explains forecast data instead of replacing the forecast model.
- Modular architecture allows APIs, AI, alerts and UI to be updated independently.
- Smart caching keeps essential information available when connectivity is weak.

HOW WE KEEP THE SYSTEM RELIABLE

- Ground every AI response in retrieved weather data and official warnings.
- Use official warning text as the fallback for critical information.
- Cache important forecasts and warnings so essential information remains accessible.
- Keep the system modular so individual components can be improved without rebuilding the whole platform.

VIABILITY

- Targets weather-sensitive users such as farmers, event planners and logistics fleets.
- The same core platform can scale to institutions and disaster-response use cases.
- API-based weather data + lightweight AI keeps the solution practical to operate.
- User feedback and verified data can continuously improve explanations and alerts.

IMPACT — WHAT CHANGES

- Faster access to understandable weather information.
- Better preparedness before severe weather events.
- Wider access for people who do not understand technical forecasts.
- Stronger disaster-management communication through timely alerts.

BENEFITS — WHAT USERS GAIN

- Time saved by getting instant answers instead of checking multiple sources.
- Lower weather-related losses for crops, logistics and events.
- Complex meteorological information made easier for non-experts.
- Forecasts, alerts and climate information brought together in one interface.

- **Goal: clearer information → earlier decisions → better preparedness**

- **References used for the conversational weather and AI approach:**

- 1 Hierarchical AI-Meteorologist: LLM-Agent System for Weather Forecast Reporting — arXiv**

<https://arxiv.org/html/2511.23387>

- 2 A Modular LLM-Agent System for Transparent Multi-Parameter Weather Interpretation — arXiv**

<https://arxiv.org/html/2512.11819>

- 3 Develop a Smart Weather Chatbot Using AutoGen**

<https://bryananthonio.com/>

- 4 Weather Chatbot — Multi-LLM Assistant (live demo)**

<https://weather-chatbot-llm.vercel.app/>